



SUBJECT: ADVANCED DATA COMMUNICATION

STUDENT ID: PUIS/24110008

STUDENT NAME: ADDOTEY VINCENT

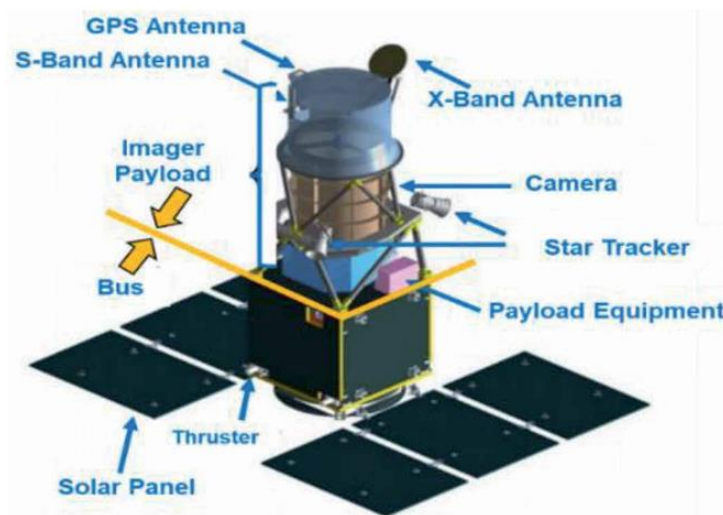
TASK DESCRIPTION

The student is required to produce a comprehensive, well-researched essay examining the architectural and functional complexities of satellite data communication. The work must demonstrate an expert-level understanding of how satellite systems facilitate global data exchange, with a specific focus on "Satellite Zooming" (High-Throughput Spot Beam technology) and high-resolution data acquisition.

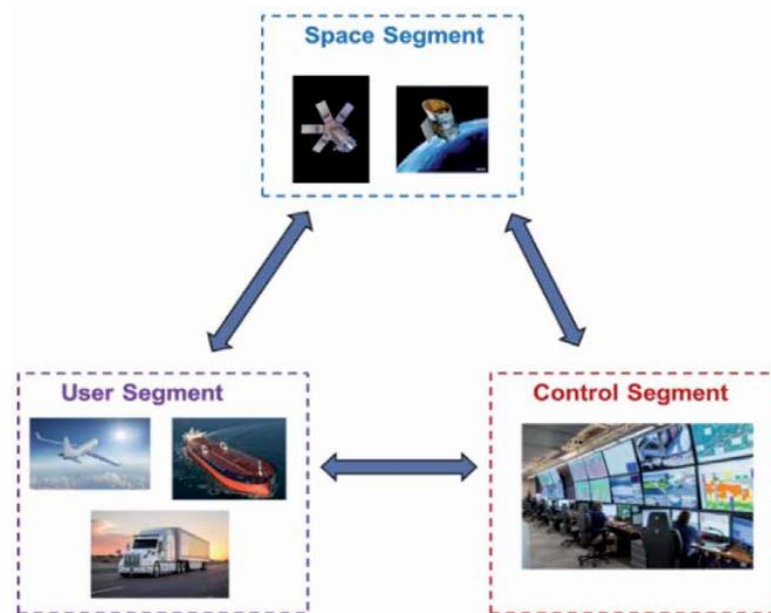
INTRODUCTION

A satellite is a spacecraft that orbits (moves around) a celestial body such as the earth. A spacecraft has several design constraints placed upon it before it can be placed in an orbit around the intended celestial body. First, satellite designs are limited in their mass and volume to fit on the launch vehicle that places them into orbit. Secondly, the mass and volume limits affect the size of the power system on the spacecraft; therefore, the amount of power available to the satellite is also limited. In addition, the space environment (thermal, radiation, atomic oxygen, space debris, micrometeoroids, etc.) imposes constraints on the design such as parts and material selection.

A spacecraft consists of two parts: the spacecraft bus and the payload (PL) [1, 2]. The spacecraft bus provides control of the satellite and support services to the mission payload, while the mission payload provides the mission part of the satellite including payload control, mission data processing, and mission data downlink dissemination. Some examples of mission payloads (or payloads or PLs) includes: scientific instruments, remote sensing instruments, navigation service transmitters, or communications equipment. A satellite may have one type of payload (PL) or a combination of payload types to accomplish its mission such as navigation, remote sensing, and communications. Shown below is a typical imaging satellite used for the remote sensing mission. The clear separation between the spacecraft bus that provides solar power and maneuvering capability via thruster, while the payload consisting of the camera and supporting communication devices such as antennas and guidance devices such as star trackers



Below is also a typical satellite with bus and payload separation



The successful development of satellite technology paved the way for a global communications satellite industry. In August 20, 1964, the world's first commercial communications satellite was formed. On April 6, 1965, the first Intelsat satellite, Early Bird (also called Intelsat 1), was launched; it was designed and built by Rosen's team at Hughes Aircraft Company. Early Bird was the first operational commercial satellite providing regular telecommunications and broadcasting services between North America and Europe. Early Bird was followed by Intelsat 2B and 2D, launched in 1967 and covering the Pacific Ocean region, and Intelsat 3 F-3, launched in 1969 and covering the Indian Ocean region. Intelsat's satellites in geostationary orbit provided nearly global coverage, as Arthur C. Clarke had envisioned 24 years earlier. Nineteen days after Intelsat 3 F-3 was placed over the Indian Ocean, the landing of the first human on the Moon on July 20, 1969, was broadcast live through the global network of Intelsat satellites to over 600 million television viewers. The Soviet Union continued its development of satellite technology with the Molniya series of satellites, which were launched in a highly elliptical orbit to enable them to reach the far northern regions of the country. The first satellite in this series, Molniya 1, was launched on April 23, 1965. By 1967 six Molniya satellites provided coverage

throughout the Soviet Union. During the 50th anniversary of the Soviet Union on October 1, 1967, the annual parade in Red Square was broadcast nationwide via the Molniya satellite network. In 1971 the Intersputnik International Organization of Space Communications was formed by several communist countries, led by the Soviet Union. The potential application of satellites for development and their ability to reach remote regions led other countries to build and operate their own national satellite systems. Canada was the first country after the Soviet Union and the United States to launch its own communications satellite, Anik 1, on November 9, 1972. This was followed by the launch of Indonesia's Palapa 1 satellite on July 8, 1976. Many other countries followed suit and launched their own satellites.



The sputnik 1 satellite

Advances in satellite technology have given rise to a healthy satellite services sector that provides various services to broadcasters, Internet service providers (ISPs), governments, the military, and other sectors. There are three types of communication services that satellites provide: **telecommunications**, **broadcasting**, and **data communications**. Telecommunication services include telephone calls and services provided to telephone companies, as well as wireless, mobile, and cellular network providers.

Broadcasting services include radio and television delivered directly to the consumer and mobile broadcasting services. DTH, or satellite television, services (such as the DirecTV and DISH Network services in the United States) are received directly by households. Cable and network programming is delivered to local stations and affiliates largely via satellite. Satellites also play an important role in delivering programming to cell phones and other mobile devices, such as personal digital assistants and laptops.

Data communications involve the transfer of data from one point to another. Corporations and organizations that require financial and other information to be exchanged between their various locations use satellites to facilitate the transfer of data through the use of very small-aperture terminal (VSAT) networks. With the growth of the Internet, a significant amount of Internet traffic goes through satellites, making ISPs one of the largest customers for satellite services. Satellite communications technology is often used during natural disasters and emergencies when land-based communication services are down. Mobile satellite equipment can be deployed to disaster areas to provide emergency communication services.

One major technical disadvantage of satellites, particularly those in geostationary orbit, is an inherent delay in transmission. While there are ways to compensate for this delay, it makes some applications that require real-time transmission and feedback, such as voice communications, not ideal for satellites. Satellites face competition from other media such as fibre optics, cable, and other land-based delivery systems such as microwaves and even power lines. The main advantage of satellites is that they can distribute signals from one point to many locations. As such, satellite technology is ideal for “point-to-multipoint” communications such as broadcasting. Satellite communication does not require massive investments on the ground making it ideal for underserved and isolated areas with dispersed populations.

In a relatively short span of time, satellite technology has developed from the experimental (Sputnik in 1957) to the sophisticated and powerful. Mega-constellations of thousands of satellites designed to bring Internet access to anywhere on Earth are in development. Future communication satellites will have more onboard processing capabilities, more power, and larger-aperture antennas that will enable satellites to handle more bandwidth. Further improvements in satellites’ propulsion and power systems will increase their service life to 20–30 years from the current 10–15 years. In addition, other technical innovations such as low-cost reusable launch vehicles are in development. With increasing video, voice, and data traffic requiring larger amounts of bandwidth, there is no dearth of emerging applications that will drive demand for the satellite services in the years to come. The demand for more bandwidth, coupled with the continuing innovation and development of satellite technology, will ensure the long-term viability of the commercial satellite industry well into the 21st century.

Satellite communication mostly dependent on ground communication systems and the parameters responsible for the spacecraft communication depends on the range, frequency, size and power of ground antennas. The command and telemetry provide constant interaction with the spacecraft [7] so the modulation scheme plays an important role in telemetry and command. The huge data which is accumulated got converted in digital bit stream so as to transmit to the ground for display. Wider channel bandwidth and higher powers to establish reliable satellite links due to non-linear distortion imposes constraint on the reliable down linking of the data. Selection of the modulation technique plays critical role and various parameters such as dynamic range, linearity, minimization of interferences are important for link margin establishment. Reliable link margin establishment for the communication with the spacecraft is calculated based on link budget. Link budget needs the following information's: • Planned data or information rate. • Modulation type • Forward error correction rate (1/2 or 3/4) • Uplink and Downlink frequencies. • Uplink and Downlink antenna sizes. • Uplink and Downlink antenna efficiency. • Uplink and Downlink transmit and receive gains at frequency. • Minimum digital signal strength (E_b/N_0) for desired Bit Error Rate (BER) performance. The main challenge in present communication system design is to avoid signal distortion, sensitivity degradation and capture signals with appropriate levels as Amplitude modulation is orthogonal to phase and frequency modulation. AM requires linearity in comparison to FM or for PM where requirement of phase linearity to be met. But interferences and thermal noise causes phase to jitter resulting in an error. BER below minimum value causes outages in link. The modulation scheme chosen for satellite communication is based on power and bandwidth efficiency. Power efficiency refers to the energy required in each bit to transmit the data at a specified BER. Channel bandwidth can be optimized by judicious selection of modulation and coding. The squeezing of maximum data into the available spectrum is the main selection criteria of the modulation scheme. The spectral efficiency parameter defined these criteria of transmission data in the assigned bandwidth defined in terms of bits per second per Hz (bps/Hz). The spectral efficiency can be enhanced by various techniques such as pre-modulation filtering, reduction in fast transition. Various filtering techniques such as raised cosine, square-root raised cosine, gaussian filtering but raised cosine filters provide better spectral efficiency. Enhancement of power results in the weight increase and lowering of power can lead to less efficient operation so the optimum solution is the selection of modulation scheme meeting all the requirements of critical components like power amplifiers. Communication satellite maximum capacity per unit weight is

desired and to achieve the same output transmitter are driven hard to overcome the thermal noise. This forces amplifiers to be driven at saturation and require modulation techniques having less non-linearities in amplitude. Amplitude modulation is orthogonal to phase and frequency modulation but AM requires linearity in comparison to FM or PM for which phase linearity is an important parameter. But interferences and thermal noise causes phase to jitter resulting in an error. So the compromise between various modulation parameters to be carried out for the selection of modulation scheme which is primarily governed by the application and in case of satellite communication adoption of digital techniques is mostly preferred.

The use of satellites in communications systems is very much a fact of everyday life, as is evidenced by the many homes equipped with antennas or “dishes” used for reception of satellite television. What may not be so well known is that satellites form an essential part of telecommunications systems worldwide, carrying large amounts of data and telephone traffic in addition to television signals. Satellites offer a number of features not readily available with other means of communications. Because very large areas of the earth are visible from a satellite, the satellite can form the star point of a communications net, simultaneously linking many users who may be widely separated geographically. The same feature enables satellites to provide communications links to remote communities in sparsely populated areas that are difficult to access by other means. Of course, satellite signals ignore political boundaries as well as geographic ones, which may or may not be a desirable feature. A good overview of the role of satellites is given by Pritchard (1984) and Brown (1981). Satellites are also used for remote sensing, examples being the detection of water pollution and the monitoring and reporting of weather conditions. Some of these remote sensing satellites also form a vital link in search and rescue operations for downed aircraft and the like. The earth station should be in a position to control the satellite if it drifts from its orbit and if subjected to any kind of drag from the external forces.

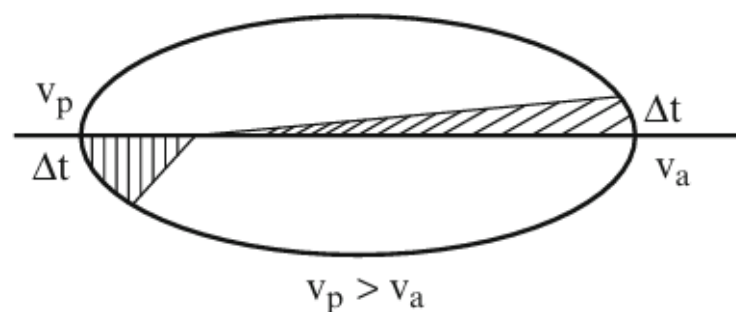
TECHNICAL PRINCIPLES: ANALYSIS OF ORBITAL MECHANICS (LEO, MEO, GEO), FREQUENCY SPECTRUM MANAGEMENT (C, KU, KA BANDS), AND SIGNAL MODULATION.

Orbital mechanics is the branch of physics that deals with the motion of objects in space under the influence of gravitational forces. It is primarily concerned with the motion of satellites,

planets, moons, and other celestial bodies in orbit around each other. The fundamental principle behind orbital mechanics is Isaac Newton's law of universal gravitation, which states that **any two objects with mass attract each other with a force that is directly proportional to the product of their masses and inversely proportional to the square of the distance between them**. This gravitational force is responsible for keeping objects in orbit around larger bodies like planets or stars. The motion of objects in orbit can be described using Kepler's laws of planetary motion, formulated by Johannes Kepler in the early 17th century. These laws provide a mathematical description of how objects move in elliptical orbits around a central mass. The three laws are as follows:

1. Kepler's First Law (the law of ellipses): Every planet or satellite moves in an elliptical orbit around the central mass (usually a star or a planet), with the central mass located at one of the foci of the ellipse.

2. Kepler's Second Law, (the law of areas): The line connecting a planet or satellite to its central mass sweeps out equal areas in equal intervals of time. This means that an object in orbit moves faster when it is closer to the central mass and slower when it is farther away.



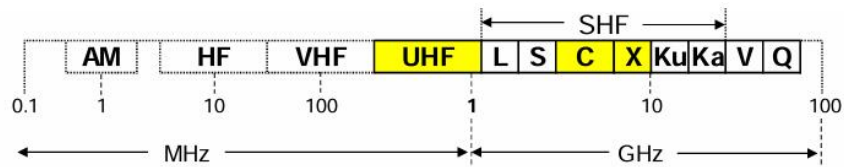
3. Kepler's Third Law, (the law of periods): The square of the orbital period of a planet or satellite is proportional to the cube of its average distance from the central mass. In other words, the farther an object is from the central mass, the longer its orbital period. These laws, along with Newton's laws of motion, form the basis for understanding and predicting the motion of objects in space. Engineers and scientists use orbital mechanics to calculate and plan satellite launches, design spacecraft trajectories, and predict celestial events such as eclipses and planetary conjunctions.

Orbital mechanics is a complex field that involves various mathematical techniques, such as differential equations and vector calculus, to solve the equations of motion. It also takes into account other factors such as atmospheric drag, solar radiation pressure, and the gravitational influence of multiple bodies when analyzing more complex orbital scenarios.

Almost every communication industry including broadcasting, navigation, transportation, and cellular relies on satellite technology in some way. Applications and services for satellite communications include broadband communications, mobile satellite services, and weather forecasting. Emerging applications and services for satellite communications include in-flight connectivity, connected car, and 5G New Radio non-terrestrial network. With strong demand for faster data throughput, satellite communications use complex modulation schemes to improve their spectral efficiency. The modulation techniques for satellite communications require not only faster data rates but also minimizing the impact of nonlinear amplification in the radio-frequency (RF) power amplifier. This white paper focuses on modulation techniques for modern satellite communications from generating and analyzing the signals to the impacts of phase noise on modulation quality.

Allocating frequencies to satellite services is a complicated process which requires international coordination and planning. This is carried out under the auspices of the International Telecommunication Union (ITU). Frequency Spectrum concepts:

- Frequency: Rate at which an electromagnetic wave reverts its polarity (oscillates) in cycles per second or Hertz (Hz).
- Wavelength: distance between wavefronts in space. Given in meters as: $\lambda = c/f$ Where: c = speed of light (3×10^8 m/s in vacuum) f = frequency in Hertz
- Frequency band: range of frequencies.
- Bandwidth: Size or “width” (in Hertz) of a frequency band.
- Electromagnetic Spectrum: full extent of all frequencies from zero to infinity.



International Telecommunication Union



Electromagnetic spectrum is the valuable but limited resource that makes possible virtually every mission that the National Aeronautics and Space Administration (NASA) undertakes, including in the areas of earth science, space science, human space exploration and aeronautical research. Yet even as spectrum access is growing increasingly important for NASA missions, the migration of the Internet from wireline platforms to the world of mobile communications is creating unprecedented pressure for more spectrum capacity to be made available for commercial wireless broadband services. What is Spectrum? The basic building block of radio communications are radio waves. The key characteristics of spectrum are the propagation features and the amount of information that signals can carry. A signal is broadly defined as a detectable quantity (e.g., current, voltage, electromagnetic field) that varies in time. An important signal for radio communications is one where the quantity varies as a periodic sine wave. Frequency is the number of cycles that occur in one second and is defined in units of Hertz (Hz), which is another name for cycle per second. The frequency of a signal is the reciprocal of its period. The amplitude of the signal is half the peak to-peak separation of the quantity that is changing with time, e.g., voltage. In general, signals sent using the higher frequencies have lower propagation distances but a higher data-carrying capacity and the opposite for longer wavelengths with lower frequencies makes up for the Radiofrequency (RF) spectrum. The range from 3 kiloHertz (kHz) to 300 GigaHertz (GHz)

that may be used for wireless communication. Shorter wavelengths with higher frequencies make up the optical spectrum, while the part of the spectrum that can be seen is the visible spectrum. NASA relies on a range of tools for communicating and creating images utilizing almost every single component of the electromagnetic spectrum in one way or another. This report is designed to provide a better understanding of basic issues for NASA spectrum access and management by addressing: what spectrum is; how it is managed; key topics with respect to international and domestic policy decisions; how NASA organizes its spectrum management activities; opportunities and challenges; and future spectrum requirements for space missions. Like all federal spectrum users, NASA requires assured access to spectrum to reflect the long-term nature of investments, in systems – including satellites – designed to remain in operation for decades, versus the 18-month lifecycle of new cellphones. Thus, new commercial technology trends present both opportunities and challenges. In addition, passive scientific use of spectrum – for example in the form of Earth remote sensing satellites that glean weather and climate information – must be protected from unwanted interference in the face of expanding activity from active services, such as wireless broadband.

Spectrum Basics: A transmitter generates a radio wave, which is then detected by a receiver. An antenna allows a transmitter to send energy into space and a receiver to pick up energy from space. iii Transmitters and receivers are typically designed to operate over a limited range of frequencies. Use of the radio spectrum is regulated, access is controlled and rules for use are enforced because of the potential for interference between uncoordinated uses. Spectrum is scarce because at any time and place, one use of a frequency precludes its use for another purpose. This has given rise to technological innovation and regulatory flexibility to increase the extent to which spectrum is shared between different users, either on the basis of time or geography.

Regulatory Environment: In the US, two main policy bodies oversee spectrum use – the Department of Commerce’s National Telecommunications and Information Administration (NTIA) and the Federal Communications Commission (FCC). For federal spectrum, NTIA is the lead authority and for non-federal spectrum, the FCC has regulatory purview. Both agencies work together to ensure that policy decisions promote spectrum use consistent with both US economic interests and government user requirements. The dual management framework used in the US is unique in the world; most other countries have a single government agency perform the spectrum management function over both public and private sector uses. US activities regarding international frequency management are carried out under the leadership of the Department of

State (DoS), whose role is to ensure the US speaks with one voice and that all applicable precedents and treaties are followed. International regulation and coordination of spectrum use occurs through the International Telecommunication Union (ITU), in compliance with a set of international rules and regulations (also known as the Radio Regulations). The ITU is the UN agency for matters dealing with information and communications technology. The State Department, along with NTIA and the FCC, oversees US preparations for ITU meetings. The ITU publishes the Radio Regulations (RR) as agreed at World Radiocommunication Conferences (WRCs), which occur every three to four years and update the RR, including spectrum allocations to various services. ITU member nations generally develop national frequency management regulations consistent with the RR. NASA Spectrum Management: Within this overarching national policy framework, NASA organizations manage the Agency's use of spectrum and ensure NASA access to spectrum for all missions using space-based and ground-based assets. The NASA Spectrum Management Program ensures that all NASA activities comply with national and international rules and regulations applicable to spectrum use and facilitates securing spectrum and orbital resources (both domestically and internationally) to enable aeronautical and space mission requirements. Agency spectrum management responsibilities reside in the Space Communications and Navigation (SCaN) program office. SCaN's Deputy Associate Administrator oversees the overall planning, policy development, and administration of the NASA Spectrum Management Program. The Director of Spectrum Policy and Planning oversees: NASA activities to obtain adequate international and national allocations of frequency bands; to pursue coordination of proposed NASA frequency use with other domestic and foreign users of the frequencies; and to obtain approval of frequency assignments by NTIA. The National and International Spectrum Program Managers implement spectrum management initiatives and day-to-day spectrum management activities. They represent NASA in national and international fora, including committees and working groups responsible for developing frequency allocations and associated technical and procedural rules. Center Spectrum Managers provide the latest communications requirements regarding implementation of projects and missions at the Center to the National and International Spectrum Program Managers. NASA has adopted procedures for requesting frequency assignments and obtaining new frequency allocations in order to effectively implement national and international spectrum management policy. In almost all cases, identification of spectrum support for NASA needs is focused on frequency bands allocated nationally and internationally

for the particular radio service for which the Agency requires support. This includes both terrestrial use (e.g. fixed, mobile, radiolocation, radionavigation and other terrestrial radio service allocations) and space use (e.g., space radio service allocations that support the US space programs). NASA Spectrum Use: NASA's investigations of the universe are accomplished mainly through the use of unmanned automated spacecraft. These missions employ both passive and active instruments operating at various frequencies to measure data from the Earth, planets, Sun, and other phenomena. They require highly reliable communications, often over long periods of time and at great distances, to guide and control the spacecraft and to bring back the images and new scientific data they collect. NASA's spacecraft must be: tracked and controlled; the health, condition, and safety of the spacecraft must be monitored; and the scientific data that is obtained must be transmitted to mission operators. Channels are needed for robust, low data rate communications designed for critical Tracking, Telemetry, and Commanding (TT&C), and in the case of crewed spacecraft, for voice services. Still other channels are needed for high data rate communications for transmission of mission data. NASA operates a sophisticated global infrastructure, consisting of several component networks, including the Deep Space Network (DSN), which is used to support scientific spacecraft; Space Network (SN), which provides tracking and data relay for spacecraft, satellites, and expendable launch vehicles using space and ground segments. It includes the Tracking and Data Relay Satellite System (TDRSS); and Near Earth Network (NEN), which consists of ground stations owned by NASA, commercial entities, and other partners that provide communications and tracking services to missions in the near earth region. Earth science research at NASA has turned space-based observing technology and scientific expertise to use for the study of Earth and its integrated system of land, ocean, atmosphere, ice, and biological processes. Spaceborne sensors, which rely on spectrum, are the only tools that can provide environmental data repetitively on a global scale. To support the intermediate and long term goals of understanding the Earth, NASA has developed numerous radio sensors designed to fly onboard spacecraft. NASA's space science missions are designed to study the mysteries of the universe from origins to destiny, to explore the solar system, to discover planets around other stars, to search for life beyond Earth and to chart the evolution of the universe and understand its galaxies, stars, planets, and life. NASA's scientific investigations are carried out via spectrum-dependent unmanned automated spacecraft. v NASA efforts to open the frontiers of space by human exploration could not be accomplished without spectrum. In exploring space,

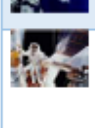
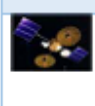
NASA has brought together a wide range of technologies, machines and people in the development of such programs as the Space Transportation System (STS) and the International Space Station (ISS). NASA is developing the capabilities needed to send humans to an asteroid by 2025 and Mars in the 2030s. Yet another spectrum-reliant focal point for NASA initiatives is Aeronautics and Space Technology Development. NASA is focused on developing technology solutions that will eliminate barriers to growth for the civil aviation system and provide a safe, efficient national aviation system.

National Spectrum Issues: Key drivers of the explosive growth in mobile broadband include the development of smartphones and other mobile computing devices, the emergence of broad new classes of connected devices (the Internet of things) and the rollout of Fourth-Generation (4G) wireless technologies such as Long Term Evolution (LTE). Several portions of the radio spectrum that are under consideration to keep up with wireless industry spectrum demand, including for unlicensed devices, carry potential implications for NASA equities. To meet this burgeoning commercial demand, in 2010, the FCC released a National Broadband Plan that called for making additional spectrum available for wireless broadband use – 500 MegaHertz (MHz) by 2020, setting off a national spectrum “policy” race for even more spectrum bands to be considered to meet this goal. In a June 2010 Executive Memorandum, the President adopted this 500 MHz goal as US policy. A 2013 Executive Memo focused on increased sharing. Sharing efforts were further promoted when, in 2012, the President’s Council of Advisors on Science and Technology (PCAST) concluded that the traditional method for accommodating new services by clearing and reallocating portions of federal spectrum is not a sustainable model for national spectrum policy because it is too costly and takes too much time. PCAST made a series of recommendations for federal spectrum focused on large-scale sharing opportunities. NASA’s spectrum use is also affected by the needs of the growing commercial space sector, which has grown beyond the provision of basic communications and broadcasting services through satellites, to the provision of Earth resources information and development of new markets such as space tourism. The private sector is developing launch vehicles and facilities, as well as payloads, which require access to spectrum resources. To support the increased spectrum requirements arising from the growth of this sector, NASA is working with federal regulators to assess ways that federal spectrum traditionally used by NASA for its own operations can be made available, when needed, by private sector stakeholders for commercial launches and operations in space.

International Issues: Because NASA missions require spectrum access on a global basis, NASA must influence

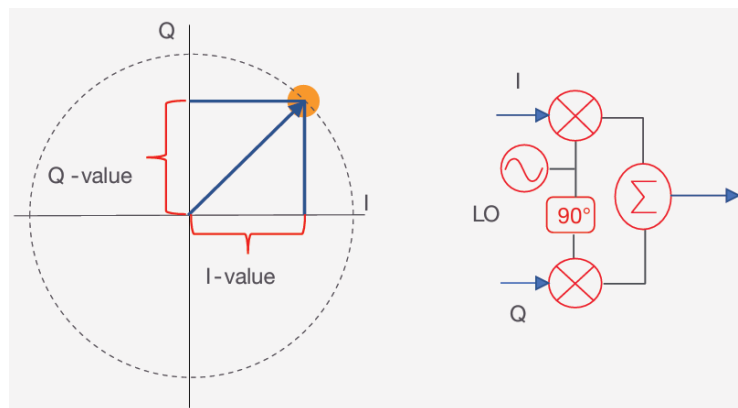
developments in the international arena to ensure that its capabilities that are spectrum-dependent will have access to spectrum where and when needed. There are two paths within the US for the development of spectrum management rules and procedures contained in the international RR. One is a technical path where engineering studies concerning radio matters are conducted, which provide technical bases for international RR; the other is a more political path where preparations leading to a WRC involve discussions in domestic and international bodies. NASA participates in both domestic and international forums in making the technical studies necessary to promote and protect operations in the space science services. It also participates in bilateral and multilateral discussions, where negotiations occur that could affect these operations. As with domestic spectrum policy trends, WRC issues in recent conference cycles have faced pressures relating to the dynamic tension between public sector users and commercial mobile wireless providers around the globe. Often, there is an interplay between domestic and international policy directions, including an interest on the part of US technology developers in the identification of certain bands for mobile broadband applications as part of the WRC process to help create further leverage for such bands being put into play for commercial reallocation at home (for example the 5 GHz bands.). At WRC-12, NASA succeeded in getting several items of specific interest included for consideration on its agenda and subsequently approved at WRC-15,: (1) A possible Earth Exploration Satellite Service (EESS) uplink allocation in the 7-8 GHz range to help alleviate congestion in the 2 GHz bands; (2) A review of current limits on use of 410-420 MHz for Space Research Service (SRS) operations; (3) An expansion of 600 MHz around 9.5 GHz to support EESS (active) operations; and (4) Consideration of nanosat/picosat regulatory issues. While NASA remains vigilant about a range of potential WRC issues that could affect its operations, there are several spectrum access issues that raise potentially higher concerns. These include: (1) spectrum allocations for mobile broadband/International Mobile Telecommunication (IMT); (2) increasing broadband spectrum for the fixed satellite service; and (3) various proposals for spectrum in or adjacent to passive sensing bands. Conclusion: The spectrum management environment, which continues to be shaped by advanced technology capabilities, evolving policy frameworks and emerging operating requirements, poses both opportunities and challenges for NASA spectrum equities. NASA remains committed to proactively addressing both to meet its own mission requirements and help to contribute to balanced policy decisions regarding spectrum access. US policymakers are confronting what industry has described as a spectrum capacity crunch, with

several bands that are under study for future wireless industry use carrying potential implications for NASA equities. At the ITU, the ITU-Radiocommunications Sector (ITU-R) is addressing future demand challenges, as well, as it works to finalize its vision of the connected future via planning efforts for Fifth-Generation (5G) Wireless. Simply put, challenges and opportunities in this complex electromagnetic spectrum environment are redefining how NASA carries out its roles as a proactive spectrum steward.

	Band	Frequency Range
	Very Low Frequency (VLF) Dynamic Spectrograph of a VLF "Whistler".	(3 kHz-30 kHz)
	Low Frequency (LF) STEREO/WAVES (SWAVES) uses LF radio imaging to study coronal mass ejections.	(30 kHz-300 kHz)
	Medium Frequency (MF) Non-Directional Beacons use MF radio frequencies to deliver directional information to aircraft. NASA and the Federal Aviation Administration (FAA) are working to improve non-directional beacon technology.	(300 kHz-3000 kHz)
	HF Also called "short wave" and used for long distance communications; NASA uses HF at test ranges.	(3 MHz-30 MHz)
	VHF NASA utilizes the VHF band by airborne sensors to study the thickness of sea ice.	(30 MHz-300 MHz)
	UHF NASA astronauts use UHF systems as backups for their voice communication systems. NASA also uses these frequencies to remotely sense the Earth's surface (e.g. soil moisture with the Soil Moisture Active Passive (SMAP) mission) and atmosphere.	(300 MHz-3000 MHz)
	Super High Frequency (SHF) Numerous NASA and National Oceanic and Atmospheric Administration (NOAA) missions use SHF frequencies to probe the Earth's atmosphere and surface. The Tropical Rainfall Measuring Mission and Global Precipitation Measurement missions observe the structure of rain in 3 dimensions.	(3 GHz-30 GHz)
	Tracking and Data Relay Satellite (TDRS) The current configuration consists of nine in-orbit satellites (four first generation, three second generation and two third generation satellites) distributed to provide near continuous information relay service to missions like the ISS.	S band (2-4 GHz) Ku-Band (12-18 GHz) Ka-band (27-40 GHz)
	Extremely High Frequency (EHF) Numerous NASA and NOAA missions can use EHF frequencies to probe the Earth's surface, although they primarily are used to probe the atmosphere.	(30GHz-300GHz)
	Infrared (IR) The Geostationary Operational Environmental Satellite (GOES) satellites use IR technology to view and track hurricane paths.	(.003 x 10 ¹⁴ Hz-4 x 10 ¹⁴ Hz)
	Visible Terra uses the visible light spectrum to take true color images of the Earth and its features.	(4 x 10 ¹⁴ Hz-7.5 x 10 ¹⁴ Hz)
	Ultraviolet (UV) The Solar and Heliospheric Observatory is studying the sun- from its core to its outer corona to its outer wind.	(7.5 x 10 ¹⁴ Hz-3 x 10 ¹⁶ Hz)
	X-Ray The GOES satellites use an X-ray imager to photograph and study the sun.	(3 x 10 ¹⁶ Hz- Upward)
	Gamma Ray/Cosmic Ray Gamma Rays or Cosmic Rays are used to further study the universe.	

Modulation Schemes for Modern Satellite

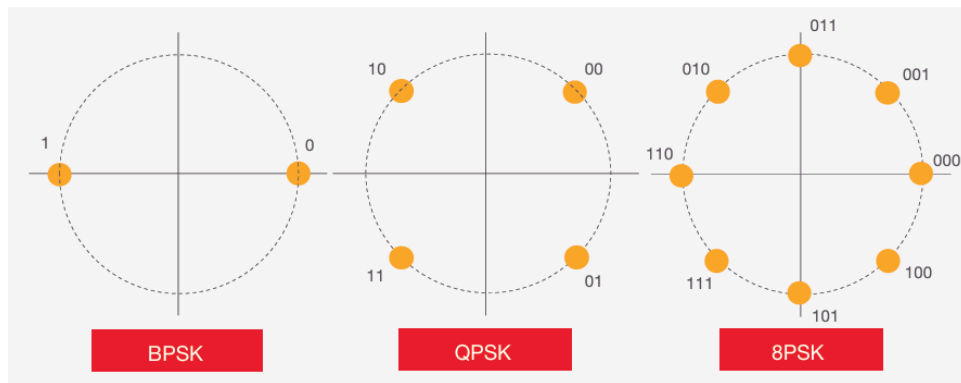
Communications Amplitude, frequency, and phase are basic modulation methods that apply to a carrier signal. You can express modulating signals in polar form (vector) as a magnitude and phase. Digital modulation is often expressed in terms of I (in-phase) and Q (quadrature) components by removing carrier frequency. In the picture below, I and Q signals mix with the same local oscillator (LO) but with a 90-degree phase shifter placed in one of the LO paths. The main advantage of I/Q modulation is the symmetric ease of combining independent signal components into a single composite signal for transmitting, then splitting that composite signal into its separate components for receiving



There are two main categories of modulation schemes: constant envelope and nonconstant envelope. “Constant envelope” means that all constellation points have a fixed distance from the center.

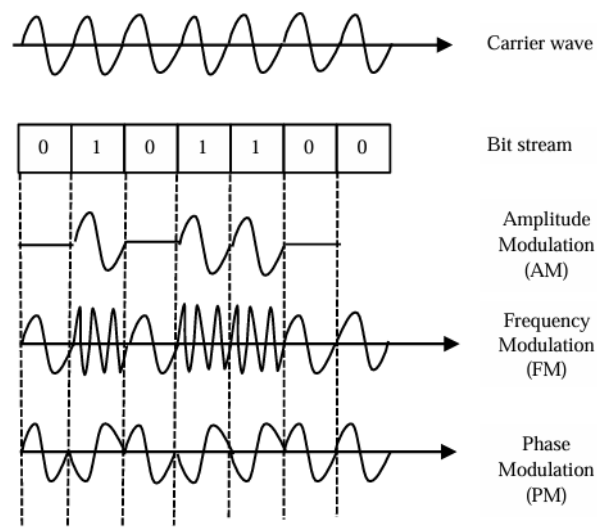
Constant envelope digital modulation schemes:

The constant envelope modulation schemes are the most suitable for satellite communications because they minimize the effect of nonlinear amplification in the high-power amplifier. Those schemes include frequency-shift keying (FSK) and phase-shift keying (PSK). To achieve faster data rates, higher-order modulation schemes provide better spectral efficiency, but they are more sensitive to channel impairments. The picture below is an illustration of the constellation diagrams of binary PSK (BPSK), quadrature PSK (QPSK), and 8PSK. They transmit 1, 2, and 3 bits per symbol, correspondingly. For higher-order PSK, the constellation points are closer to each other, and the system is more sensitive to channel impairments.



Radio frequency links are employed in the satellite operations such as controlling, receiving, health monitoring, spacecraft parameters checking, rate ranging, payload operations in the form of telecommand and telemetry signals which are perceived to be complex and are having restricted information. Various modulation and multiplexing techniques in advanced communication systems are employed keeping into considerations of various parameters such as data rate, BER, spectral efficiency, susceptibility to interference etc. Particularly modulation techniques are critical and employed for securing the data and enabling to establish reliable link margin and highly efficient on-board system. In the present scenario, analog modulation schemes are replaced with the digital based techniques where digital bits represent information. Analog systems employed in the past are prone to performance variations due to routing and bunching of wires, device placement, interconnections, improper grounding and shielding along with various radiation effects encountered due to closed loop formation. Digital based technology overcomes the bottlenecks faced in the analog based system [1]-[2] mainly usage of large spectrum for transmit/receive information whereas digital systems are more efficient. based on maximizing data rate and minimizing transmitted power, channel bandwidth, symbol error apart from resistance to interference signal. Present day satellite communication uses number of antenna beams which in turn increases interferences arising from frequency reuse. The methodology to overcome this challenge is to employ digital modulation with coding together with on board demodulation and re-modulation to separate uplink and down link signal degradation. Digital modulation and demodulation techniques demand increase complexity and high level of integration to transfer large amount of data and information. Excessive filtering to improve spectral efficiency can reduce interferences but increases BER (bit error rate) due to blurring of transmitted symbols. This also

contributes in making the system bulkier and increases the footprint. The main parameters in analog modulation is characterized with the parameter SNR (signal-to-noise ratio) whereas in digital modulation is characterized by BER (bit error rate). The requirement for minimum BER for voice is 10^{-3} and for data 10^{-6} . The selection of the particular modulation scheme involves consideration of several requirements of the system such as BER, power consumption, linearity, bandwidth, reliability and complexity factor apart from weight and cost.



Satellites carry several transponders having defined bandwidth resulting in the fixed data rate which is determined by the modulation scheme and access method. The future satellites may require a large number of commands to be uplinked in a relatively short time. This may call for a command system which can handle higher bit-rates. Digital modulation scheme can be classified as constant envelope such as FSK, PSK and non constant envelope such as ASK, QAM. M-APK (Amplitude and phase keying) and n-m PCM are the other choices. FSK is having lower bandwidth efficiency due to requirement of higher bandwidth for data modulation whereas QAM requires very high C/N ratio and is not useful for satellite communication. Mostly constant envelope scheme is preferred for satellite communication [4]-[5] and major parameters are BER, spectral efficiency and power efficiency. QPSK is the accepted modulation technique for low to medium bit rate transmission. For bulk data transmission, high density spectrum efficient modulations

techniques are to be employed. Due to trade-off between bandwidth and carrier power, 4 ϕ (PSK) is having widespread usage compared to FM and QPSK signals in communication systems. PSK modulation is widely used in satellite data transmission taking into consideration the spectral efficiency offered by the scheme at a given transmitted power. BPSK, QPSK, OQPSK, MSK have inherent 3-dB advantage over coherently detected FSK technique. Further QPSK is preferred than BPSK due to doubling of data rate as carriers are in phase quadrature [6]. Also symbol energy of QPSK is more compared to the bit energy in BPSK. The data rate of a QPSK modulator is twice the data rate of I and Q data streams but the error probability is higher compared to the BPSK. MSK is a type of PSK requiring higher bandwidth and complex circuit. This article details various modulation schemes, their selection criteria, concept of BER, and implementation methodologies.

SELECTION OF MODULATION SCHEME:

Selection of digital modulation scheme is based on maximizing data rate and resistance to interference along with minimizing various parameters such as channel bandwidth, probability of symbol error, transmitted power and current. There are three basic types of modulation methods for transmission of digital signals. These methods are based on the three attributes of a sinusoidal signal, amplitude, frequency and phase. The corresponding modulation methods are called Amplitude Shift Keying (ASK), Frequency shift keying (FSK), and Phase Shift Keying (PSK). The selection is based on the criteria of minimizing non-linearities and interferences for signal detection. In addition, combination of ASK and PSK is employed at higher bit rates which is known as Quadrature Amplitude Modulation (QAM) Modulation signals with constant envelope are required for reducing side-lobe regeneration [8]. The scheme that generates less amplitude fluctuation to avoid signal fidelity is an important consideration for aerospace applications (Table.1). As the selection is based on power and band-width efficiency which is governed by the E_b/N_0 criteria where bit error probability (BER) of $< 10^{-5}$ is taken as reference.

AI-DRIVEN APPROACHES FOR REAL-TIME SATELLITE DATA PROCESSING AND ANALYSIS

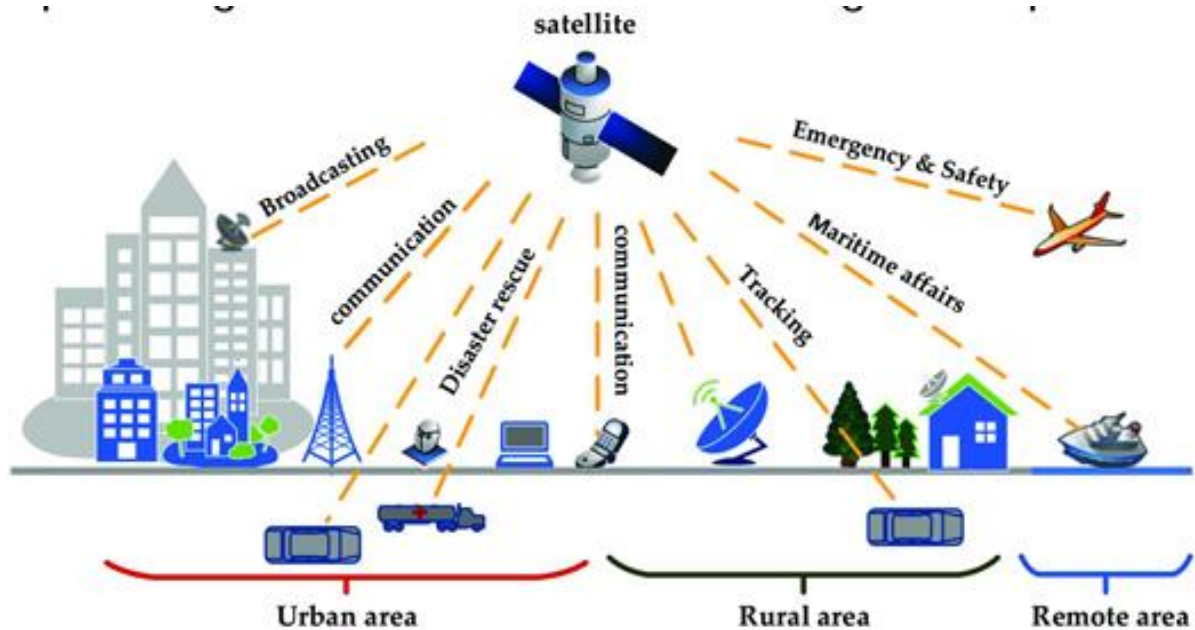
Integrating Artificial Intelligence (AI) into satellite data processing significantly advances Earth science by enabling real-time analysis of vast and complex datasets. AI-driven approaches, utilizing machine learning (ML) and deep learning (DL) techniques, enhance the efficiency and accuracy of data interpretation, crucial for disaster response, climate monitoring, and precision

agriculture. Advanced AI methods, such as reinforcement learning and generative adversarial networks (GANs), offer innovative solutions for handling diverse satellite data, optimizing observation timing, and generating synthetic data to fill coverage gaps. Onboard AI systems further boost real-time processing by analyzing data as it is collected, reducing latency and bandwidth usage, which is vital for rapid disaster assessment and response. Integrating data from multiple sources, including drones, ground sensors, and crowdsourced information, enhances the comprehensiveness and accuracy of Earth observation systems. However, challenges in AI deployment remain, including data availability, training optimization, data quality management, and uncertainty handling. Future work should focus on developing advanced AI techniques, improving model interpretability, and establishing robust ethical and governance frameworks. Collaborative efforts among AI researchers, Earth scientists, government agencies, and commercial entities are essential to drive innovation and ensure responsible AI use in Earth science. These advancements will significantly improve our ability to monitor and manage Earth's systems, providing critical insights for immediate and long-term decision-making

REAL-TIME SATELLITE DATA PROCESSING

The Real-time satellite data processing is crucial for timely decision-making in applications such as disaster response, environmental monitoring, and precision agriculture. AI significantly enhances this process by handling large datasets efficiently, improving data accuracy, and enabling onboard processing for immediate analysis.

1. **Timely Decision Making:** Immediate analysis and response to changing condition
2. **Disaster Response:** Providing early warnings and helping in the coordination of emergency services during natural disasters such as hurricanes, floods, and wildfires. AI analyzes real-time satellite images to quickly detect and assess damage from events like hurricanes and floods, enabling faster and more effective emergency response. In agriculture, AI processes satellite imagery to monitor crop health, predict yields, and detect soil moisture levels, providing farmers with actionable insights to optimize resource use and increase productivity.



3. **Environment Monitoring:** Deforestation, ice melt, and pollution levels
4. **Agriculture Efficiency:** Crop health, soil conditions, and weather patterns
5. **National Security:** Timely intelligence and situational awareness
6. **Urban Planning and Infrastructure:** Current data on land use, traffic patterns, and urban growth.

CORE AI TECHNOLOGIES

Artificial Intelligence encompasses a range of technologies designed to process large datasets and perform complex computations, enhancing the capabilities of real-time satellite data processing. The comparison between ground processing and on-board processing

1. **Supervised Learning:** Involves training models on labeled data to make predictions or classify new data. Common algorithms include Random Forest, Support Vector Machines (SVM), and Neural Networks.
2. **Unsupervised Learning:** Analyzes and clusters data without pre-labeled outputs. Techniques like K-means clustering help in identifying patterns and anomalies in satellite data.
3. **Deep learning Convolutional Neural Networks (CNNs):** Particularly effective for image recognition and classification tasks, such as identifying specific features or changes in satellite images over time. **Recurrent Neural Networks (RNNs):** Suitable for time-series

analysis and sequential data, making them ideal for tracking changes and trends in satellite data over time

CHALLENGES AND FUTURE WORKS

1. **Data Availability:** Securing ample, diverse, and high-resolution datasets for AI model training remains a challenge, especially in remote or underexplored regions.
2. **Training Optimization:** Achieving effective learning from vast and intricate Earth science data demands substantial computational resources and expertise in hyperparameter tuning.
3. **Data Quality:** Fluctuations in data quality, like inconsistencies or noise in satellite images, can compromise the accuracy and dependability of AI predictions.
4. **Uncertainty:** Accurately managing and quantifying uncertainty in AI model predictions is imperative for making dependable scientific inferences and informed decisions.
5. **Model Interpretability:** Ensuring AI models are interpretable and comprehensible to scientists and stakeholders is crucial for fostering trust and usability.
6. **Diversity:** Integrating various data types (e.g., diverse sensor modalities) and enabling AI models to generalize across different conditions and regions pose complex challenges.
7. **Integrity and Security:** Safeguarding AI model integrity and data security against tampering or cyber-attacks is vital for upholding trustworthiness and precision.

Artificial Intelligence (AI) has emerged as a transformative solution to address these challenges. AI techniques such as Machine Learning (ML), Deep Learning (DL), Neural Networks, Computer Vision, and Natural Language Processing (NLP) enable satellites to make intelligent decisions, detect patterns, and analyze data in real time. By embedding AI capabilities in satellites, engineers can achieve greater autonomy, reliability, and operational efficiency. AI-based models can learn from past data, detect anomalies, optimize communication links, and predict hardware failures before they occur, ensuring mission success and cost efficiency. In satellite communication, AI plays a vital role in signal optimization, dynamic bandwidth allocation, and interference management, especially in the era of Low Earth Orbit (LEO) satellite constellations. These constellations, consisting of hundreds or even thousands of interconnected satellites, require intelligent coordination to manage communication traffic and avoid collisions. AI helps automate

such processes by predicting orbital changes, optimizing routing paths, and improving quality of service (QoS) for global Internet coverage. In Earth observation and remote sensing, AI-powered satellites use image recognition and deep learning algorithms to process high-resolution images, identify land-use changes, monitor deforestation, track weather patterns, and assess disaster impacts. This enables faster and more accurate decision-making for governments, industries, and researchers. Additionally, AI models are being trained to detect space debris, predict its trajectory, and assist in collision avoidance, which is becoming increasingly critical due to the growing congestion in Earth's orbit.

FUTURE ADVANCEMENTS. By integrating advanced AI techniques, such as reinforcement learning and GANs, with multi-source data integration, Earth observation systems will become more accurate and comprehensive. Improved model interpretability will enhance trust among stakeholders, while real-time processing capabilities onboard satellites will enable immediate responses in disaster management and environmental monitoring. Collaboration and ethical governance will be essential to ensure responsible and effective AI implementation in Earth science. The methodology of satellite technology integrated with artificial intelligence involves a systematic approach that enhances the capability of satellites to collect, process, and interpret data intelligently. In this approach, satellites equipped with advanced sensors continuously capture data from the Earth's surface, atmosphere, and surrounding space environment. The collected data may include optical images, radar signals, and environmental readings such as temperature, humidity, and pollution levels. Once the data is gathered, it undergoes preprocessing to remove noise, correct distortions, and standardize formats so that it becomes suitable for analysis. After preprocessing, the data is transmitted to ground stations or cloud-based systems where artificial intelligence algorithms are applied for interpretation. Machine learning and deep learning techniques play a major role in this stage. These AI models are trained on large datasets to recognize patterns, detect anomalies, and make predictions. For example, convolutional neural networks can analyze satellite images to identify deforestation, urban expansion, or changes in agricultural patterns. Similarly, predictive algorithms can help forecast weather conditions or detect natural disasters like floods and cyclones by analyzing atmospheric data. Artificial intelligence also supports the autonomous

functioning of satellites themselves. Through reinforcement learning and adaptive control systems, satellites can optimize their orbits, manage power consumption, and schedule communication tasks without continuous human intervention. The integration of AI allows satellites to make decisions in real time, improving response speed and accuracy while reducing dependence on ground control. The methodology further includes a continuous feedback process in which AI models are refined over time. The system compares the outcomes of predictions with real-world results, allowing it to learn from errors and improve future accuracy. In advanced systems, AI models are even deployed onboard satellites, enabling real-time edge processing directly in space. This reduces data transmission delays and allows immediate analysis of critical information. Overall, the methodology emphasizes a data-driven and learning-based process that transforms traditional satellite systems into intelligent and autonomous platforms. By combining artificial intelligence with satellite technology, the efficiency of data processing, environmental monitoring, disaster management, and communication networks is significantly improved, leading to faster decision-making and more reliable global observations.

The fusion of Artificial Intelligence (AI) and satellite technology is still in its early stages, with immense potential to revolutionize space systems in the coming decades. Building upon the findings of this research, the future scope for AI in satellite technology can be summarized as follows: Future satellites are expected to evolve into fully autonomous agents capable of self-management without continuous ground control. AI algorithms combined with advanced edge computing will enable satellites to make real-time decisions regarding navigation, communication, power management, and mission adjustments. This will reduce operational costs and increase mission efficiency. With continued advancements in AI hardware such as low-power GPUs, Tensor Processing Units (TPUs), and neuromorphic processors, onboard AI processing will become increasingly feasible. Edge AI will allow satellites to perform complex image analysis, anomaly detection, and decision-making without dependence on ground station processing, significantly reducing latency and bandwidth usage.

THE ROLE OF AI IN SATELLITE DATA PROCESSING

1. **Handling Large Datasets:** AI algorithms excel at processing vast amounts of satellite data quickly and accurately.
2. **Enhancing Data Accuracy:** Machine learning models can improve the accuracy of data classification and anomaly detection, reducing false positives and negatives.
3. **Predictive Analytics:** AI techniques such as predictive modeling help forecast weather patterns, environmental changes, and potential disasters.
4. **Automation and Efficiency:** AI automates repetitive tasks such as data cleaning, pre-processing, and initial analysis, freeing up human resources for more complex tasks.
5. **Onboard Processing:** AI enables onboard data processing on satellites, reducing the need for extensive data transmission to ground stations.
6. **Data Fusion:** AI integrates data from multiple satellite sources, creating a comprehensive and cohesive dataset that improves analysis and decision-making.
7. **Improved Data Visualization:** Advanced AI-driven visualization tools help in interpreting complex data, making it easier for stakeholders to understand and act upon the information.

Conclusion

The integration of Artificial Intelligence (AI) with satellite technology represents a transformative advancement in the field of space science and engineering. This research demonstrates that AI-enabled satellite systems significantly enhance autonomy, efficiency, and accuracy across various domains, including Earth observation, communication optimization, predictive maintenance, and disaster monitoring. AI models such as Convolutional Neural Networks (CNNs) achieved high accuracy in satellite image classification, enabling efficient environmental monitoring and resource mapping. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models effectively forecast satellite subsystem failures, supporting proactive maintenance and reducing downtime. Reinforcement learning algorithms dynamically adjust communication parameters, resulting in reduced latency, higher bandwidth utilization, and improved signal reliability. AI integration significantly reduces detection and response times for natural disasters, highlighting its potential in emergency management and humanitarian aid. These advancements highlight that AI can transform satellites from passive data collectors into intelligent, autonomous agents capable of real-time decision-making and adaptive behavior. However, challenges such as

computational constraints, data quality issues, cybersecurity risks, and ethical considerations must be addressed to ensure sustainable and secure deployment of AI in space systems. The future of satellite technology lies in the synergy between AI, edge computing, and advanced hardware innovations. In conclusion, AI integration offers a paradigm shift in satellite operations, promising unprecedented improvements in performance, reliability, and mission capability. Continued research, international collaboration, and investment in AI-driven space technologies will be essential to fully realize this vision, paving the way for the next generation of autonomous satellite networks that can operate efficiently and intelligently in the complex environment of space.

REFERENCES

1. AI-Driven Approaches for Real-Time Satellite Data Processing and Analysis Hafez Ahmad¹, and Shakila Islam Jhara² ¹Mississippi State University, and ²University of Chittagong
2. Mathieu DERVIN Brussels, 6th May 2015 Evolution of Satellite Communication Systems
3. Satellite Communications-Dennis Roddy, 2ndEdition, 1996, McGraw Hill.
4. Satellite Communications: Design Principles –M. Richcharia, 2ndEd., BSP, 2003.
5. Digital Satellite Communications –Tri. T. Ha, 2ndEd., MGH, 1990.
6. Fundamentals of Satellite Communications –K. N. Raja Rao, PHI, 2004.
7. Satellite Communications –Timothy Pratt, Charles Bostian, Jeremy Allnutt, 2ndEdition, 2003, John Wiley & Sons.
8. Satellite Communications Engineering –Wilbur, L. Pritchard, Robert A. Nelson and Heur